Deep Learning HW #3

For this homework assignment, we read Deep Face Recognition by Parkhi et al. from the University of Oxford. This paper had two main goals, the first which is to propose a technique to build a large dataset of human faces which requires minimal human power for annotation. They used this procedure to develop a dataset of over 2 million faces that balances scale and purity. The second goal of the paper is to test multiple different CNN architectures for face recognition, and discover a much simpler, but deep, network can achieve almost state-of-the-art results on face recognition benchmarks.

The five-stage, semi-automatic pipeline for building a face dataset starts with a candidate list of public figures, the pipeline expands images via search engines, filters out noise automatically, removes duplicates, then performs a final light-touch human pass where annotators validate blocks of ranked images rather than labeling each image.

Stage	Aim	Type	# of persons	# of images per person	total # images	Annotation effort	100%-EER
1	Candidate list generation	A	5,000	200	1,000,000	–	–
2	Image set expansion	M	2,622	2,000	5,244,000	4 days	–
3	Rank image sets	A	2,622	1,000	2,622,000	–	96.90
4	Near dup. removal	A	2,622	623	1,635,159	–	–
5	Final manual filtering	M	2,622	375	982,803	10 days	92.83

(1) Choose ~5K popular identities; keep those whose first 200 downloaded images are ~90% correct; remove overlaps with LFW/YTF. (2) Expand to ~2,000 images/identity using Google/Bing queries. (3) Train one-vs-rest linear SVMs (Fisher Vector Faces) to rank and keep top 1,000 per identity for precision. (4) Remove exact/near duplicates using VLAD descriptors + tight clustering. (5) Train a multi-way CNN to rank images and let humans accept/reject blocks (targeting >95% purity), yielding ~982k labeled face images across 2,622 identities.

In their experiments, they tested a specific style of CNN, a "very deep" VGG-style blocks of small convolutions + ReLu + pooling.

layer	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
type	input	conv	relu	conv	relu	mpool	conv	relu	conv	relu	mpool	conv	relu	conv	relu	conv	relu	mpool	conv
name	–	conv1_1	relu1_1	conv1_2	relu1_2	pool1	conv2_1	relu2_1	conv2_2	relu2_2	pool2	conv3_1	relu3_1	conv3_2	relu3_2	conv3_3	relu3_3	pool3	conv4_1
support	–	3	1	3	1	2	3	1	3	1	2	3	1	3	1	3	1	2	3
filt dim	–	3	–	64	–	–	64	–	128	–	–	128	–	256	–	256	–	–	256
num filts	–	64	–	64	–	–	128	–	128	–	–	256	–	256	–	256	–	–	512
stride	–	1	1	1	1	2	1	1	1	1	2	1	1	1	1	1	1	2	1
pad	–	1	0	1	0	0	1	0	1	0	0	1	0	1	0	1	0	0	1

layer	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
type	relu	conv	relu	conv	relu	mpool	conv	relu	conv	relu	conv	relu	mpool	conv	relu	conv	relu	conv	softmx
name	relu4_1	conv4_2	relu4_2	conv4_3	relu4_3	pool4	conv5_1	relu5_1	conv5_2	relu5_2	conv5_3	relu5_3	pool5	fc6	relu6	fc7	relu7	fc8	prob
support	1	3	1	3	1	2	3	1	3	1	3	1	2	7	1	1	1	1	1
filt dim	–	512	–	512	–	–	512	–	512	–	512	–	–	512	–	4096	–	4096	–
num filts	–	512	–	512	–	–	512	–	512	–	512	–	–	4096	–	4096	–	2622	–
stride	1	1	1	1	1	2	1	1	1	1	1	1	2	1	1	1	1	1	1
pad	0	1	0	1	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0

Table 3: **Network configuration.** Details of the face CNN configuration A. The FC layers are listed as "convolution" as they are a special case of convolution (see Section 4.3). For each convolution layer, the filter size, number of filters, stride and padding are indicated.

They tested three configurations of this network, the one on table 3 Is network A. Network B, and C are the same architecture, but the difference is that B adds 2 additional convolutional layers and C adds 5. Training begins with an N-way identity classifier for N=2,622 identities. After convergence, the final classifier is replaced with a learned projection to an L-dimensional descriptor space (L=1,024), and the network is optimized for verification using a triplet-loss objective: for each triplet (anchor a, positive p, negative n), the embedding should satisfy d(a,p) + margin < d(a,n).

The ablation analysis shows that (1) more data (even with some label noise) can beat a smaller curated set, (2) 2D alignment helps primarily at test time, and (3) triplet-loss embedding learning provides the biggest accuracy lift in the unrestricted LFW protocol.

No.	Method	Images	Networks	Acc.
1	Fisher Vector Faces [21]	-	-	93.10
2	DeepFace [29]	4M	3	97.35
3	Fusion [30]	500M	5	98.37
4	DeepID-2,3		200	99.47
5	FaceNet [17]	200M	1	98.87
6	FaceNet [17] + Alignment	200M	1	99.63
7	Ours	2.6M	1	98.95

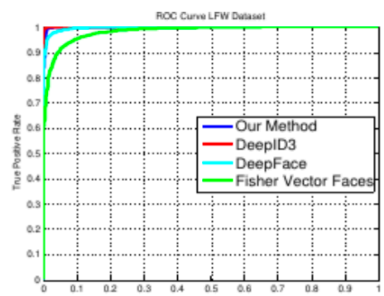

Table 5 + ROC (paper): LFW comparison and ROC curves.

No.	Method	Images	Networks	100%- EER	Acc.
1	Video Fisher Vector Faces [15]	-	-	87.7	83.8
2	DeepFace [29]	4M	1	91.4	91.4
3	DeepID-2,2+,3		200	-	93.2
4	FaceNet [17] + Alignment	200M	1	-	95.1
5	Ours ($K = 100$)	2.6M	1	92.8	91.6
6	Ours ($K = 100$) + Embedding learning	2.6M	1	97.4	97.3

Table 6 (paper): YTF results. Using K=100 faces/video and embedding learning improves performance substantially.

The key takeaways are that straightforward deep CNN trained first for classification then refined with triplet-loss metric learning can achieve strong face verification without requiring a large ensemble or 3D alignment.